

Matlab code :

```
%Converting the transfer function to state space system
```

```
num3 = [1];
```

```
den3 = [1 9 8 7];
```

```
disp('State space system=')
```

```
[A3,B3,C3,D3] = tf2ss(num3,den3)
```

Command Window Output

State space system=

A3 =

-9	-8	-7
1	0	0
0	1	0

B3 =

1
0
0

C3 =

0	0	1
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D3 =

0
